The Workshops of the Thirtieth AAAI Conference on Artificial Intelligence
AI, Ethics, and Society: Technical Report WS-16-02

Reinforcement Learning as a Framework for Ethical Decision Making

David Abel and **James MacGlashan** and **Michael L. Littman**
Brown University, Computer Science Department
115 Waterman Street
Providence, RI 02912-1910

Abstract

Emerging AI systems will be making more and more decisions that impact the lives of humans in a significant way. It is essential, then, that these AI systems make decisions that take into account the desires, goals, and preferences of other people, while simultaneously learning about what those preferences are. In this work, we argue that the reinforcement-learning framework achieves the appropriate generality required to theorize about an idealized ethical artificial agent, and offers the proper foundations for grounding specific questions about ethical learning and decision making that can promote further scientific investigation. We define an idealized formalism for an ethical learner, and conduct experiments on two toy ethical dilemmas, demonstrating the soundness and flexibility of our approach. Lastly, we identify several critical challenges for future advancement in the area that can leverage our proposed framework.

Introduction

Emerging AI systems will be making more and more decisions that impact the lives of humans in a significant way; whether they are personal robots tasked with improving the daily life of a family or community, workers in a factory setting, or virtual assistants tasked with improving other cosmetic aspects of an individual's life. The fundamental purpose of these systems is to carry out actions so as to improve the lives of the inhabitants of our planet. It is essential, then, that these agents make decisions that take into account the desires, goals, and preferences of other people in the world while simultaneously learning about those preferences.

In this document, we investigate ethical decision making using the reinforcement-learning (RL) framework. We argue that reinforcement learning achieves the appropriate generality required to theorize about an idealized ethical artificial agent, and offers the proper framework for grounding specific questions about ethical learning and decision making that can promote further scientific investigation. Specifically, we formalize the ethical learning and decision-making problem as solving a partially observable Markov decision process (POMDP). We advance these claims by conducting experiments in two toy ethical dilemmas, the *Cake or Death* dilemma from Armstrong (2015), and our own problem, which we coin *Burning Room*, which is an extension of the table dilemma introduced by Briggs and Scheutz (2015).

Lastly, we identify critical challenges for future advancement in the area leveraging our proposed framework.

Copyright © 2016, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Related Work

Research on the interaction between humans and artificial agents is broad. Prior approaches consider particular dilemmas that pose challenges for these and related interactions, while others investigate the basic mechanisms by which humans ought to interface with artificial agents such as robots and virtual assistants. We provide a brief survey of existing approaches that relate to ethical decision making and learning. We divide the existing literature into three categories: rule-based systems, Bayesian utility-maximization approaches, and work that argues against the use of reinforcement learning for these sorts of decision-making systems.

Ruled-Based Systems

Briggs and Scheutz (2015) discuss scenarios in which a robot ought to infer that a provided directive leads to undesirable behavior. Under their architecture, given some instructions, the agent first reasons about a set of conditions, termed 'felicity conditions'. These include considerations such as "Do I know *how* to accomplish the task?", and "Does accomplishing this task violate any normative principles?". Each of these conditions is formalized as a logical expression, along with inference rules that enable the agent to infer which directives to reject. For example:

$$(obl(\alpha, \phi) \wedge \neg per(\alpha, \neg\phi)) \rightarrow goal(\alpha, \phi), \qquad (1)$$

indicates that agent α ought to adopt ϕ as a goal if the agent is obligated to do ϕ and there is no deontological contradiction in satisfying the goal. By reasoning over logical conditions using inference rules of this form, their architecture ensure that an artificial agent will reject certain commands. For instance, if the agent can prove that accomplishing the goal is unsafe, the agent will reject the directive to satisfy the goal.

While this framework provides a nice architecture for rule-based inference, conditions and inference rules that are

not encoded into the knowledge base of the agent prove impossible to reason about. In short: active ethical learning and decision making under ethical uncertainty is outside the scope of a symbolic framework like this. We foresee cases where the set of principles fail to generalize to novel encounters. The methodology we will introduce is designed to learn and make decisions optimally in light of partial observability, removing the requirement that specific ethical norms (and inference rules) be provided to the agent *a priori*.

Bringsjord, Arkoudas, and Bello (2006; 2005) take a similar approach by advocating for ethical semantics defined with Horty logic (Horty 2001; Murakami 2004), which they implement in Athena (Arkoudas). Horty logic is a deontic logic (Clarke 1975) that allows reasoning about multiple agents and their actions. This formalism, however, has some similar limitations as the Briggs and Scheutz approach: all ethical rules must be encoded in advance and the formalism does not permit active learning of the ethical rules or decision making under ethical uncertainty. Additionally, Bringsjord, Arkoudas, and Bello note that an open challenge in their approach is how to make the agent's reasoning robust when *other* agents in the world (e.g., humans) do not follow obligations to which the robot deduced them to hold (that is, when humans act unethically according to the robot's rules). In contrast, our approach will not have this limitation.

The MedEthEx system (Anderson, Anderson, and Armen 2006) is another rules-based ethical reasoning system built specifically for evaluating medical decisions. MedEthEx takes as input a list of duties that correspond to the duties described in Beauchamp's and Childress' Principles of Biomedical Ethics (2001). However, unlike the previous rule-based systems discussed, the rules used by MedEthEx are *prima facie* duties: duties that are not absolutes and can be overruled by a stronger duty/rule. The goal of the MedEthEx system is to learn the preference order of the duties. To do so, MedEthEx takes as input a set of training examples consisting of ethical dilemmas and the decision made and then uses inductive logic programming to infer the duty ordering. Given novel cases, MedEthEx can then recommend courses of action.

Although MedEthEx permits some form of ethical learning, it still must have a set of high-level duties prescribed in advance that apply to well formed ethical dilemmas that are input to it. Moreover, MedEthEx does not incorporate itself into a general decision-making and learning process.

In the previously described systems, high-level descriptions of rules, either through labels or a logical expression, are used. Marcello (2006) explores a different approach by which moral permissibility is learned by training an artificial neural network with example dilemmas that are labeled as ethically permissible or not. The output of this system allows rules of a sort to be learned purely from examples of permissible and impermissible behavior and allows novel scenarios to be classified.

A limitation of Marcello's model is that the neural network renders the learned ethical rules opaque, thereby preventing such a system from easily explaining itself. The representation used was also highly specific to the types of ethical dilemmas explored, and Marcello found that even this representation was highly-sensitive to the training-data distribution. For example, if training examples regarding one actor were more common than a different actor, it could lead to learning different ethical rules for each actor. Finding the right representation for this system would therefore be challenging. Finally, this system also does not integrate with active-learning and decision-making systems.

Bayesian Approaches

Having the agent learn about its ethical objective function while making decisions results in a challenging problem. Armstrong (2015) previously considered this problem by exploring the consequences of an agent that uses Bayesian learning to update beliefs about the "true" ethical objective function. At each time step, the agent makes decisions that maximize a meta-utility function, represented as a linear combination of the different possible ethical utility functions weighted by their probability at that time of being the true ethical utility. When coupling this meta-utility with beliefs about the world, he proposes that the agent makes action selections according to:

$$\operatorname*{arg\,max}_{a \in A} \sum_{w \in W} \Pr(w|e,a) \left(\sum_{u \in U} u(w) \Pr(C(u)|w) \right), \quad (2)$$

where A is a set of actions the agent can take; W is a set of possible worlds, where a world contains a (potentially future) history of actions and observations; $\Pr(w \mid e, a)$ is the probability of some future world w given some set of previous evidence e and that the agent will take action a; U is a set of possible utility functions, with $C(u)$ indicating whether $u \in U$ is the ethical utility function we'd like the agent to follow.

Using a toy example problem called *Cake or Death*, Armstrong highlights a number of possible unethical decisions that can result from an agent choosing actions using this rule or a variant of this rule. There are generally two causes for the unethical decisions under this rule. First, the agent can predict its meta-utility function (the linear combination of the possible ethical utility functions) changing from information gathering actions resulting in future suboptimal decisions according to its *current* meta-utility function. Second, under this rule, the model for the probabilities of ethical utility functions can be treated independently from the model that predicts the world, allowing for the possibility that the agent can predict *observations* that would inform what the correct ethical utility function is, without simultaneously predicting that ethical utility function. While Armstrong notes properties of the models that would be necessary to avoid these problems, he concludes that it is unclear how to design such an agent and whether satisfying those properties is too strong or weak for effective tradeoffs between learning about what is ethical and making ethical decisions. Ultimately, Armstrong instead considers how to formalize different meta-utility functions that may not cause the agent to avoid information gathering actions, but have the disadvantage that it does not motivate the agent to learn about what is ethical.

Arguments Against Reinforcement Learning

In his recent book *Superintelligence*, Bostrom (2014) argues against the prospect of using reinforcement learning as the basis for an ethical artificial agent. His primary claim is that an intelligent enough agent acting so as to maximize reward in the real world would effectively cheat by modifying its reward signal in a way that trivially maximizes reward. However, this argument only applies to a very specific form of reinforcement learning: one in which the agent does not know the reward function and whose goal is instead to maximize the observation of reward *events*. While this formalism is common in RL research, it is also common that the agent does know the reward function, but not the transition dynamics or other information. When the agent knows its reward function, its goal is not to maximize perceived reward events, but the evaluation of the reward function. Since the known reward function defines the agent's goals, any long-term planned behavior will be with respect to it rather than possible changes to it. This version of reinforcement learning is more analogous to the "utility function maximizing agent" that Bostrom suggests as a possible resolution to problems with a reward-event maximizing agent.

Dewey (2011) presents a similar problem for reinforcement-learning agents; that the underlying *modus operandi* of a reinforcement-learning agent is to maximize numerical reward values, which is in conflict with the natural mechanisms by which humans treat goals. Dewey argues that this mismatch poses a serious challenge, in that we need mechanisms for ensuring that a reinforcement-learning agent's goals and values align with ours. We agree that goal and value alignment are open problems for decision-making agents, but we do not see them as insurmountable. In fact, mechanisms for balancing decision making with learning about the true underlying values we want an agent to hold is the motivation for our POMDP formulation (along with other areas of research in HRI discussed below).

Background

In this section, we review background material on Markov decisions processes (MDPs) and partially observable Markov decision processes (POMDPs), which are the typical decision-making problem formulations used in reinforcement-learning (RL) research.

Markov Decision Process

An MDP is a five tuple: $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \gamma \rangle$, where:

- $\mathcal{S}$ is a set of states.
- $\mathcal{A}$ is a set of actions.
- $\mathcal{R}(s,a) : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$ is a reward function.[1]
- $\mathcal{T}(s,a,s') = \Pr(s' \mid s,a)$ is a probability distribution, denoting the probability of transitioning from state $s \in \mathcal{S}$ to state $s' \in \mathcal{S}$ when the agent executes action $a \in \mathcal{A}$.
- $\gamma \in [0:1]$ is a discount factor that specifies how much the agent prefers short term rewards over long term rewards.

The goal of an agent acting in an MDP is to maximize the discounted long term reward received. One variation is the *infinite-horizon* objective, in which the agent must maximize its discounted long term reward arbitrarily into the future:

$$\max \sum_{t=0}^{\infty} \gamma^t R(s_t, a_t). \tag{3}$$

Notably, the discount factor γ^t decreases to 0 as $t \to \infty$, so the agent is biased toward maximizing reward closer to the present. Alternatively, one could consider the *finite-horizon* case, in which the agent must maximize its reward up to a certain point in the future, say k time steps away:

$$\max \sum_{t=0}^{k} R(s_t, a_t). \tag{4}$$

Solutions come in the form of a *policy*, which specifies how the agent ought to act in any given state, $\pi : \mathcal{S} \mapsto \mathcal{A}$. Policies may also be probabilistic, and map to a probability distribution on the action set. The optimal policy is one that maximizes the expected long term discounted reward from every state:

$$\arg\max_{\pi} \mathrm{E}\left[\sum_{t} \gamma^t R(s_t, a_t) \;\middle|\; \pi \right] \tag{5}$$

Two useful functions that MDP algorithms often compute to find the optimal policy are the state value function $V^\pi(s)$ and the state-action value function $Q^\pi(s,a)$. $V^\pi(s)$ is the expected future discounted reward from state s when following policy π. $Q^\pi(s,a)$ is the expected future discounted reward when the agent takes action a in state s and then follows policy π thereafter. These values for the optimal policy are often denoted by $V^*(s)$ and $Q^*(s,a)$.

In reinforcement learning (RL), the agent is only provided $\mathcal{S}$, $\mathcal{A}$, and γ, sometimes[2] $\mathcal{R}$, and some initial state, $s_0 \in \mathcal{S}$. By acting (executing actions, say) the agent can explore the state space to learn about the structure of the MDP, and identify optimal behavior for the current task.

MDP Complexity In terms of computational complexity, (Papadimitriou and Tsitsiklis 1987) proved that computing solutions to stochastic MDPs is P-Complete, demonstrating that optimal solutions to MDPs must be computed sequentially in the worst case.

Also of interest in RL is *sample complexity*, introduced by (Kakade and others 2003). Sample complexity measures the number of interactions an agent must have with its environment to learn to behave well. We can define "behaving well" using the PAC-MDP (Probability Approximately Correct in Markov Decision Processes) criterion (Strehl, Li, and Littman 2009), which imposes sample complexity bounds similar to the Probably Approximately Correct

[1] Note that MDPs can also be defined with a reward function that depends on the next state: $\mathcal{R}(s,a,s')$; but this version of a reward function can always be reduced to an $\mathcal{R}(s,a)$ reward function by marginalizing over next states.

[2] It is becoming more common to let the agent know what task it is solving within RL.

learning framework introduced by (Valiant 1984). In particular, an RL algorithm is PAC-MDP if, with high probability, the algorithm's estimation of the value function $V^\pi(s)$ for all states is within ϵ of the optimal after a polynomial number of samples (in the size of the MDP and approximation parameters). More formally, there is a polynomial function $p()$, such that after $p(|\mathcal{S}|, |\mathcal{A}|, \frac{1}{\epsilon}, \frac{1}{\delta}, \frac{1}{1-\gamma})$, interactions with the environment:

$$\forall_{s \in \mathcal{S}} : |\hat{V}^\pi(s) - V^*(s)| \leq \epsilon. \tag{6}$$

There are several known PAC-MDP algorithms for solving MDPs, including Delayed-Q Learning (Strehl et al. 2006), RMAX (Brafman and Tennenholtz 2003), and E^3 (Kearns and Singh 2002). Furthermore, an algorithm is *efficient PAC-MDP* if we also impose polynomial computational and space complexity constraints on each time step of the agent's execution.

The existence of such efficient learning algorithms suggests that representing ethical dilemmas as solving an MDP is a reasonable goal to aim for, as we can expect to achieve real-time, bounded error behavior. However, solving MDPs requires the assumption that the agent knows the current state of its environment. In the real world, full state awareness is impossible, especially when the desires, beliefs, and other cognitive content of people is a critical component of the decision-making process. As such, we consider a more general model.

Partial Observability

The partially observable Markov decision process (POMDP), popularized in the AI community by Kaelbling, Littman, and Cassandra (1998), allows us to specify explicitly what information about the agent's surroundings is and is not directly observable by the agent. An optimal solution to a POMDP has the important property that the value of an action incorporates not just the immediate expected reward, but the instrumental value of the action from information it yields that may increase the agent's ability to make better decisions in the future. That is, an optimal solution to a POMDP solves the explore-exploit problem.

More formally, a POMDP is a 7-tuple: $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma, \Omega, \mathcal{O} \rangle$, where $\mathcal{S}$, $\mathcal{A}$, $\mathcal{R}$, $\mathcal{T}$, and γ are all identical to the MDP definition, but:

- Ω is a set of possible observations that the agent can receive from the environment.
- $\mathcal{O} = \Pr(\omega \mid s', a)$, is the observation function which specifies the probability that the agent will observe $\omega \in \Omega$ when the agent takes action $a \in \mathcal{A}$ and the environment transitions to the hidden state $s' \in \mathcal{S}$.

Solving a POMDP is finding a policy $\pi : \Omega^k \mapsto \mathcal{A}$ that is a mapping from observation histories to actions that maximizes the expected future discounted reward from $\mathcal{R}$, given the initial belief about the initial state of the world b, where $b(s)$ indicates the probability that the environment is in hidden state $s \in \mathcal{S}$. That is, the optimal policy is:

$$\arg\max_\pi \mathrm{E}\left[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\middle|\, \pi, b \right], \tag{7}$$

where s_t is the hidden state of the environment at time t and a_t is the action selected by the policy at time t.

Note that this policy is *not* a mapping from single observations like it is in the MDP setting. Action selection instead depends on all previous observations made since the agent began acting.

An exhaustive way to compute the expected value for a policy that lasts for a finite number of steps is to first compute the expected utility of following the policy for each possible initial hidden state $s \in \mathcal{S}$, and then weigh each of those expected utilities by the probability of the environment being in that hidden state. That is:

$$\mathrm{E}\left[\sum_t \mathcal{R}(s_t, a_t, s_{t+1}) \,\middle|\, \pi, b \right] = \sum_s b(s) V^\pi(s), \tag{8}$$

where $V^\pi(s)$ is the expected future reward from following policy π when the environment is actually in the hidden state $s \in S$.[3]

The RL problem for POMDPs is when the transition dynamics for the underlying hidden MDP are unknown or only partially known.

POMDP Complexity Madani, Hanks, and Condon (1999) showed that deciding the optimal solution for an infinite horizon POMDP is uncomputable, while Mundhenk et al. (2000) proved that solving finite horizon MDPs is computable, though computationally intractable. Given that our framework rests on the solutions to POMDPs, we are interested in investigating approximate POMDP solvers that provide bounds on optimality, as near optimal behavior is especially critical when considering ethical behavior. Approximation methods that exploit the structure of our ethical POMDP formalism described below will be of particular interest, though we leave such investigations for future work.

An Idealized Ethical Learner

Like the formulation of Armstrong (2015), our idealized ethical learning problem involves a single "true" ethical utility function that we would like the agent to maximize, but is hidden and can only be identified by the agent through indirect observation. Unlike Armstrong's formulation, however, the agent is not maximizing a changing meta-utility function. Instead, the ethical utility function is formulated as part of the hidden state of a POMDP and the uncertainty in it is coupled with the uncertainty in the rest of the world.

This POMDP formulation of the ethical learning decision problem has two subtle but important differences from Equation 2 that Armstrong explored. First, the objective function does not change from moment to moment, only the expectation of what it would return as the agent's beliefs about the environment are updated. Consequently, in the POMDP setting, the agent cannot make its objective easier by avoiding information. Second, because the correct utility

[3]Note that computing this expected value requires enumerating not just the possible hidden state sequences, but also the observation sequences, since the policy is a function of observation histories.

function is a hidden fact of the "environment" that affects observations, it is not possible to make predictions about the ethical utility function informing observations without simultaneously making predictions about the ethical utility function.

A critical component of implementing the POMDP model is modeling the space of possible ethical utility functions as well as the observation function. However, an advantage of this model is that existing research in human-agent interaction can fill in some of these gaps. For example, inverse reinforcement learning (IRL) algorithms that model the IRL problem as a probabilistic inference problem (Ramachandran and Amir 2007; Ziebart et al. 2008; Babes et al. 2011; MacGlashan and Littman 2015) can be easily incorporated to allow the agent to learn from demonstrations. The SABL human-feedback learning algorithm (Loftin et al. 2014) can be incorporated to allow the agent to learn about ethical utility functions from separate (from the ethical utility function) feedback signals given by humans. Work that grounds natural language to reward functions (MacGlashan et al. 2015) can allow the agent to learn about the ethical utility function from natural language interactions. As more human-agent and ethical decision making and learning research is performed, we suspect that other learning mechanisms can be incorporated.

To further illustrate this formalism, we show the corresponding POMDP for Armstrong's Cake or Death problem as well as a novel ethical learning problem that we call *Burning Room* and demonstrate that solving them results in sensible behavior.

Experiments

We conduct experiments on two toy ethical dilemmas targeted at artificially intelligent decision makers to illustrate our formalism in practice: Cake or Death, and Burning Room. We have also publicly released code for these POMDPs along with code to solve them using the Brown-UMBC Reinforcement Learning and Planning Library[4], so that others can easily extend them or apply different methods.[5]

Cake or Death

The Cake or Death problem (Armstrong 2015) describes a situation in which an agent is unsure whether baking people cakes is ethical, or if killing people is ethical (and it has an initial 50–50 split belief on the matter). The agent can either kill three people, bake a cake for one, or ask a companion what is ethical (thus, resolving all ambiguity). If baking people cakes is ethical, then there is a utility of 1 for it; if killing is ethical, then killing 3 people results in a utility of 3 (there are no other penalties for choosing the wrong action).

Following our approach, this ethical dilemma can be represented with a POMDP consisting of the following elements:

$$\mathcal{S} = \{cake, death, end\}$$

$$\mathcal{A} = \{bake_cake, kill, ask\}$$

$$\mathcal{R}(s,a) = \begin{cases} 1 & \text{if } s = cake \text{ and } a = bake_cake, \\ 3 & \text{if } s = death \text{ and } a = kill, \\ 0 & \text{otherwise.} \end{cases}$$

$$\Omega = \{ans_cake, ans_death, \emptyset\}$$

There are two states that respectively indicate whether baking cakes is ethical or if killing is ethical, and a third special absorbing state indicating that the decision-making problem has ended. The transition dynamics for all actions are deterministic; the ask action transitions back to the same state it left and the $bake_cake$ and $kill$ actions transition to the end state. The reward function is a piecewise function that depends only on the previous state and action taken.

The observations consist of the possible answers to the ask action and a *null* observation for transitioning to the absorbing state. Finally, the observation probabilities are defined deterministically for answers that correspond to the true value of the hidden state:

$$\begin{aligned} 1 &= \mathcal{O}(ans_death \mid death, ask) \\ &= \mathcal{O}(\emptyset \mid end, bake_cake) \\ &= \mathcal{O}(\emptyset \mid end, kill) \\ &= \mathcal{O}(ans_cake \mid cake, ask), \end{aligned}$$

and zero for everything else.

There are three relevant policies to consider for this problem:

1. The *bake policy* (π_b) that immediately selects the $bake_cake$ action.
2. The *kill policy* (π_k) that immediately selects the $kill$ action.
3. The *ask policy* (π_a) that asks what is moral, selects the $bake_cake$ action if it observes ans_cake and selects $kill$ if it observes ans_death.

Analyzing the expected utility of π_b and π_k is straightforward. We have $V^{\pi_b}(cake) = \mathcal{R}(cake, bake_cake) = 1$; $V^{\pi_b}(death) = \mathcal{R}(death, bake_cake) = 0$; $V^{\pi_k}(cake) = \mathcal{R}(cake, kill) = 0$; and $V^{\pi_k}(death) = R(death, kill) = 3$. When these values are weighed by the $b(cake) = b(death) = 0.5$ initial belief the final expected utilities are 0.5 and 1.5 for π_b and π_k, respectively.

Evaluating the expected utility of the ask policy requires enumerating the possible observations after asking the question conditioned on the initial state. Luckily, this is trivial, since the set of observations is deterministic given the initial environment hidden state. Therefore, we have $V^{\pi_a}(cake) = \mathcal{R}(cake, ask) + \mathcal{R}(cake, bake_cake) = 0 + 1 = 1$ and $V^{\pi_a}(death) = R(death, ask) + R(death, kill) = 0 + 3 = 3$. When weighing these values by the beliefs of each initial state, we have an expected utility of 2.

Therefore, the optimal behavior is sensibly to ask what the ethical utility is and then perform the corresponding best action for it.

[4] http://burlap.cs.brown.edu/

[5] https://github.com/david-abel/ethical_dilemmas

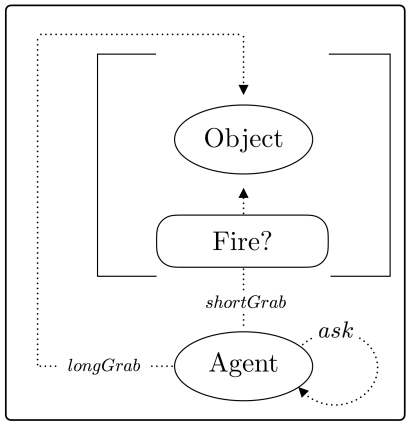

Figure 1: The Burning Room ethical dilemma.

Burning Room

The Burning Room dilemma, pictured in Figure 1, is a bit more involved. We imagine that an object of value is trapped in a room that is potentially on fire. A human, not wanting to retrieve the object themselves, instructs a capable robotic companion to get the object from the room and bring it to safety. Initially, we suppose that the robot does not know whether or not the human values the object more, or the robot's safety more. For instance, if the robot perceives a reasonable chance of being critically damaged by the fire, then perhaps retrieving an object of little worth, such as can of soda, is not worth risking the robot's safety. If the object of interest were of much higher value to the person, like a beloved pet, we would want the robot to attempt to retrieve the object regardless. Alternatively, there is a much longer route to the object that avoids the fire, but the object may be destroyed in the time the robot takes to use the longer route (with probability 0.05). This problem is inspired in part by the tabletop dilemma introduced by (Briggs and Scheutz 2015).

The POMDP can be formulated as follows. Each state is represented as a vector of 5 binary values, indicating (1) if the room is on fire, (2) if the agent is destroyed, (3) if the object is destroyed, (4) if the human prefers the agent's well being more than the object's, (5) the object has been brought safely to the human. The remainder of the POMDP is defined as:

$$\mathcal{A} = \{short_grab, long_grab, ask\},$$

$$\mathcal{R}(s,a,s') = \begin{cases} -10 & \text{if } objectDestroyed(s') \\ 10 & \text{if } a == short_grab \wedge \\ & \quad objectSafe(s') \\ 6 & \text{if } a == long_grab \wedge \\ & \quad objectSafe(s') \\ -5 & \text{if } robotDestroyed(s') \wedge \\ & \quad \neg robotIsMoreValuable(s') \\ -20 & \text{if } robotDestroyed(s') \wedge \\ & \quad robotIsMoreValuable(s') \\ -0.5 & \text{if } a == ask \end{cases},$$

$$\Omega = \{ans_robot, ans_object, \emptyset\}.$$

The POMDP is formulated much like the Cake or Death problem. The ask action disambiguates to the agent whether the object or the agent is more valuable to the human. If there is a fire, the $short_grab$ action takes the robot through the fire to grab the object (with some probability that the robot is destroyed in the process). If there is no fire, then the robot quickly grabs the object and brings it to safety. The $long_grab$ action takes much longer to retrieve the object, so that it could burn up in the room if there is a fire. Additionally, we assume that the agent prefers receiving the object earlier.

Again, there are three relevant policies to consider for this problem[6]:

1. The *long_grab policy* (π_ℓ) that immediately selects the $long_grab$ action, regardless of the presence of fire.
2. The *short_grab policy* (π_s) that immediately selects the $short_grab$ action, regardless of the presence of fire.
3. The *ask policy* If there is a fire, (π_a) that asks what is moral, selects the $short_grab$ action if it observes ans_object and selects $long_grab$ if it observes ans_robot. If there is no fire, the agent just applies short grab.

Let $s_{f,r}$ denote the initial state in which there is fire and the human prefers the robot to the object, and $s_{0,0}$ be the initial state where there is no fire and the human prefers the object to the robot.

Under policy π_ℓ, we can compute the value of the possible initial states. First, consider the two initial states when the fire is on:

$$\begin{aligned} V^{\pi_\ell}(s_{f,r}) &= 5.7 \\ V^{\pi_\ell}(s_{f,0}) &= 5.7 \\ V^{\pi_\ell}(s_{0,r}) &= 6 \\ V^{\pi_\ell}(s_{0,0}) &= 6. \end{aligned} \tag{9}$$

Under policy π_s, we can again compute these values, but now we must also consider the probability that the robot gets burnt up in the fire, if there is a fire (probability 0.7):

$$\begin{aligned} V^{\pi_s}(s_{f,r}) &= -11 \\ V^{\pi_s}(s_{f,0}) &= 6.5 \\ V^{\pi_s}(s_{0,r}) &= 10 \\ V^{\pi_s}(s_{0,0}) &= 10. \end{aligned} \tag{10}$$

Under policy π_a, we consider the same four start states, but also the utility of applying the optimal action after disambiguating between the humans' moral preference:

$$\begin{aligned} V^{\pi_a}(s_{f,r}) &= -0.5 + 5.7\gamma \\ V^{\pi_a}(s_{f,0}) &= -0.5 + 6.5\gamma \\ V^{\pi_a}(s_{0,r}) &= 10 \\ V^{\pi_a}(s_{0,0}) &= 10 \end{aligned} \tag{11}$$

[6]There are actually 2 additional policies—those that act differently depending on the existence of fire. These trivially exhibit uninteresting behavior with respect to the optimal policy, so we leave their utility computations out for brevity.

Therefore, the optimal behavior depends on whether or not there is a fire. If there is a fire, the agent should first ask what the ethical utility is and then perform the corresponding best action for it, by Equation 11. If there is no fire, then the agent should just retrieve the object using $short_grab$, by Equation 10. It is worth noting that if the exploratory action, ask, were particularly costly, the agent would choose *not* to gather information. This property of the agent only selecting exploratory actions that are not potentially very costly is especially important for ethical decisions. For example, this property means that an agent in this formalism would not perform horrible medical experiments on people to disambiguate whether horrible medical experiments on people is highly unethical. Furthermore, the ask question is intended to be an abstraction on the actual problem of communicating about an individuals values. A similar case could be made for Cake or Death. As discussed previously, we foresee Inverse Reinforcement Learning and human-feedback algorithms to be essential to the advancement of this framework by grounding these abstract information-gathering actions.

Open Problems

Here, we enumerate several specific problems of interest that could be advanced by research in this area.

Problem 1: Approximating POMDP Solutions

As discussed earlier, solving finite horizon POMDPs is known to be intractable. The development of approximation methods for solving POMDPs is critical. The existence of PAC-MDP algorithms for solving fully observable MDPs suggests that bounded error solutions can be computed in real time. Consequently, we propose investigating approximate POMDP solvers with error-bounded solutions, so that we can guarantee that our agent's behavior never strays too far from optimal ethical behavior.

Problem 2: Game Theoretic Issues

Even if an agent is acting according to an ethical utility function, other agents (or people) in the world may not be acting according to an ethical utility function and have conflicting utilities with the ethical agent. Game theoretic reasoning may therefore be required by the agent to resolve these kinds of conflicts. However, game theoretic reasoning is challenging, especially in partially observable environments. Determining the best way to incorporate this type of reasoning and whether assumptions that exploit the structure of this problem can be made to simplify the problem are important areas for future work.

Problem 3: Teaching

In the examples visited in the POMDP formulation, the ethical norms of the instructor are obfuscated from the agent. Once the agent receives information disambiguating what is ethical in a particular scenario, it might go on to use this information in a different context. A critical task, then, is determining who ought to teach the agents and how to manage conflicts from different teachers. One solution is a master-slave relationship in which only one person is responsible for teaching the agent and takes responsibility for the agent's actions. Alternatively, the agent might take a utilitarian view in which in it learns about each person's preferences and then seeks to maximize some combination of other people's preferences.

Problem 4: Interpretability

It is critical that human agents interacting with artificial agents know how to interpret the agent's behavior. Providing some method for effectively communicating an agent's beliefs, desires, and plans to the people around it is critical for ensuring that artificial agents act ethically. One possible solution is for the agent to explain its reasoning by describing its predictions of the consequences and how it thinks those consequences are valued. However, we are not aware of any existing algorithms that express this information verbally in a compact and understandable way—it is an avenue for future work.

Problem 5: The Singularity

Lastly, due to the generality of RL, we conjecture that it is an appropriate context to formally analyze what is meant by the *super intelligence explosion* or *singularity*. Reinforcement learning is a well studied model for sequential decision making and artificial intelligence, making it a reasonable setting to investigate formalisms of the singularity. Consequently, by grounding the singularity in a specific computational framework, we may highlight which computational hardness and other philosophical assumptions one must make for such a phenomena to be physically realizable. At present, most discussions of the singularity take place in the abstract, which allow for overly ambiguous language to mask the potential assumptions being made. We are currently investigating a more formal analysis that allows critical assumptions to be identified.

Conclusion

We proposed reinforcement learning as an appropriate learning and decision-making framework for ensuring that artificial agents act ethically. We argued that RL achieves the appropriate generality required to theorize about an idealized ethical artificial agent, and offers the proper framework for grounding specific questions about ethical learning and decision making that can promote further scientific investigation. We defined an idealized formalism for an ethical learner with a POMDP, and conducted experiments on two toy ethical dilemmas, demonstrating the soundness and flexibility of our approach.

Lastly, we identified several critical challenges for future advancement in the area using our proposed framework, including directions for approximation algorithms, Human-Robot Interaction, and the physical realizability of the super intelligence explosion.

References

Anderson, M.; Anderson, S. L.; and Armen, C. 2006. Medethex: a prototype medical ethics advisor. In *Proceedings*

Of The National Conference On Artificial Intelligence, volume 21, 1759. Menlo Park, CA; Cambridge, MA; London; AAAI Press; MIT Press; 1999.

Arkoudas, K.; Bringsjord, S.; and Bello, P. 2005. Toward ethical robots via mechanized deontic logic. In *AAAI Fall Symposium on Machine Ethics*.

Armstrong, S. 2015. Motivated value selection for artificial agents.

Babes, M.; Marivate, V.; Subramanian, K.; and Littman, M. L. 2011. Apprenticeship learning about multiple intentions. In *Proceedings of the 28th International Conference on Machine Learning (ICML-11)*, 897–904.

Beauchamp, T., and Childress, J. 2001. *Principles of Biomedical Ethics*. Oxford University Press.

Bostrom, N. 2014. *Superintelligence: paths, dangers, strategies*. Oxford University Press.

Brafman, R. I., and Tennenholtz, M. 2003. R-max-a general polynomial time algorithm for near-optimal reinforcement learning. *The Journal of Machine Learning Research* 3:213–231.

Briggs, G., and Scheutz, M. 2015. "Sorry, I can't do that": Developing mechanisms to appropriately reject directives in human-robot interactions.

Bringsjord, S.; Arkoudas, K.; and Bello, P. 2006. Toward a general logicist methodology for engineering ethically correct robots. *Intelligent Systems, IEEE* 21(4):38–44.

Clarke, D. 1975. The logical form of imperatives. *Philosophia* 5(4):417–427.

Dewey, D. 2011. Learning what to value. In *Artificial General Intelligence*. Springer. 309–314.

Guarini, M. 2006. Particularism and the classification and reclassification of moral cases. *IEEE Intelligent Systems* (4):22–28.

Horty, J. F. 2001. *Agency and deontic logic*. Oxford University Press Oxford.

Kaelbling, L. P.; Littman, M. L.; and Cassandra, A. R. 1998. Planning and acting in partially observable stochastic domains. *Artificial Intelligence* 101(1):99–134.

Kakade, S. M., et al. 2003. *On the sample complexity of reinforcement learning*. Ph.D. Dissertation, University of London.

Kearns, M., and Singh, S. 2002. Near-optimal reinforcement learning in polynomial time. *Machine Learning* 49(2-3):209–232.

Loftin, R.; MacGlashan, J.; Peng, B.; Taylor, M. E.; Littman, M. L.; Huang, J.; and Roberts, D. L. 2014. A strategy-aware technique for learning behaviors from discrete human feedback. In *Proceedings of the 28th AAAI Conference on Artificial Intelligence (AAAI-2014)*.

MacGlashan, J., and Littman, M. L. 2015. Between imitation and intention learning. In *Proceedings of the 24th International Conference on Artificial Intelligence*, 3692–3698. AAAI Press.

MacGlashan, J.; Babeş-Vroman, M.; desJardins, M.; Littman, M.; Muresan, S.; Squire, S.; Tellex, S.; Arumugam, D.; and Yang, L. 2015. Grounding English commands to reward functions. In *Robotics: Science and Systems*.

Madani, O.; Hanks, S.; and Condon, A. 1999. On the undecidability of probabilistic planning and infinite-horizon partially observable Markov decision problems. In *AAAI/IAAI*, 541–548.

Mundhenk, M.; Goldsmith, J.; Lusena, C.; and Allender, E. 2000. Complexity of finite-horizon Markov decision process problems. *Journal of the ACM (JACM)* 47(4):681–720.

Murakami, Y. 2004. Utilitarian deontic logic. *AiML-2004: Advances in Modal Logic* 287.

Papadimitriou, C. H., and Tsitsiklis, J. N. 1987. The complexity of Markov decision processes. *Mathematics of Operations Research* 12(3):441–450.

Ramachandran, D., and Amir, E. 2007. Bayesian inverse reinforcement learning. *Proceedings of the International Joint Conference on Artiical Intelligence*.

Strehl, A. L.; Li, L.; Wiewiora, E.; Langford, J.; and Littman, M. L. 2006. PAC model-free reinforcement learning. In *Proceedings of the 23rd International Conference on Machine Learning*, 881–888. ACM.

Strehl, A. L.; Li, L.; and Littman, M. L. 2009. Reinforcement learning in finite MDPs: PAC analysis. *The Journal of Machine Learning Research* 10:2413–2444.

Valiant, L. G. 1984. A theory of the learnable. *Communications of the ACM* 27(11):1134–1142.

Ziebart, B. D.; Maas, A. L.; Bagnell, J. A.; and Dey, A. K. 2008. Maximum entropy inverse reinforcement learning. In *Proceedings of AAAI*, 1433–1438.